Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

OFFICIAL RULEBOOK 2026

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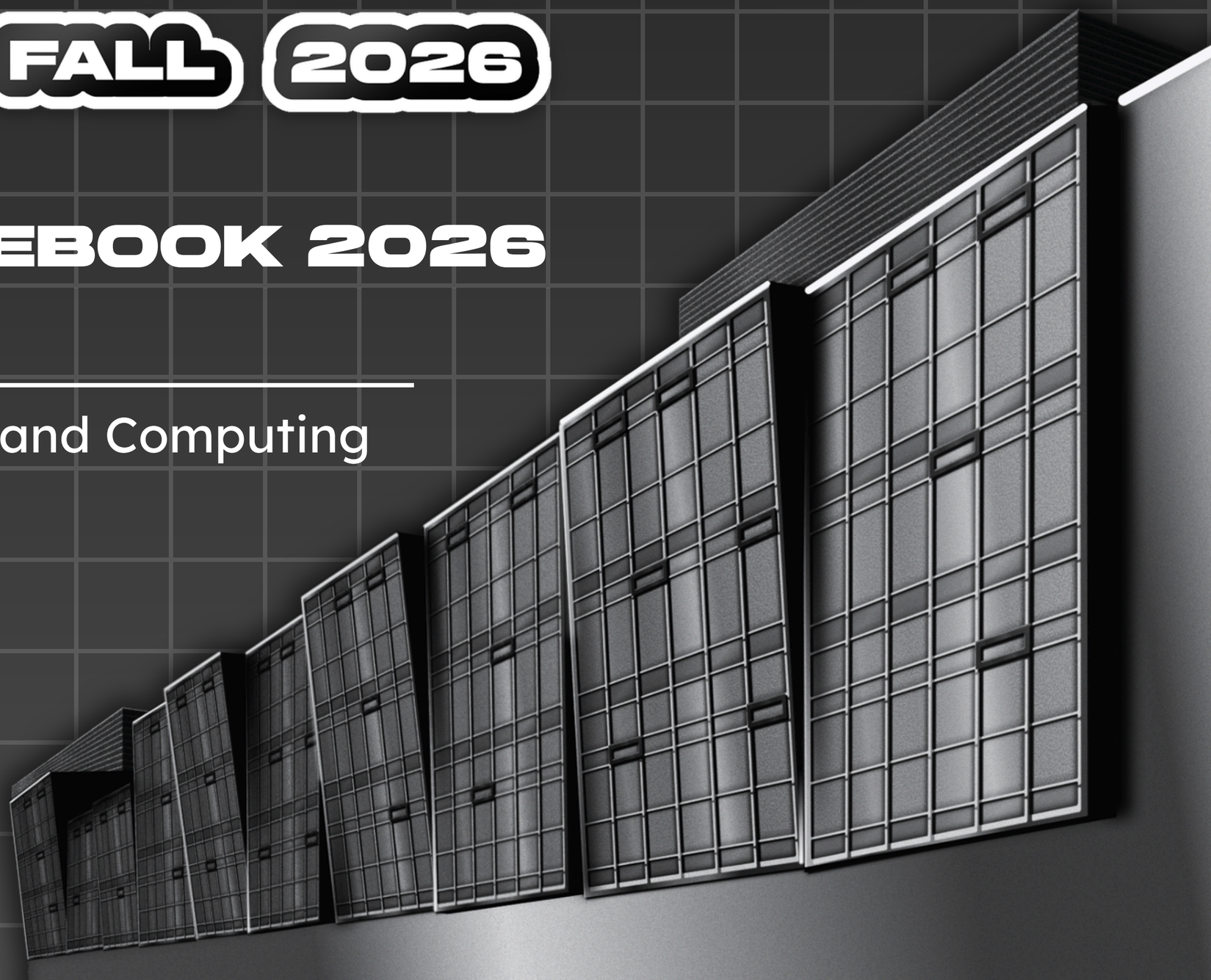
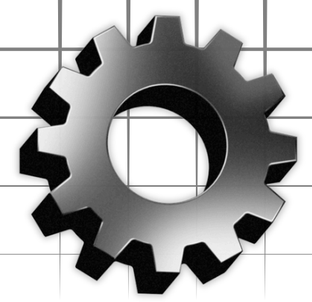


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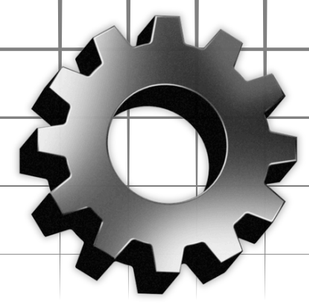
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1. OEC GENERAL RULES

1.1 Definitions

1.1.1 “Engineering Student Societies’ Council of Ontario” henceforth referred to as ESSCO, refers to the provincial association of undergraduate engineering student societies in Ontario.

1.1.2 “Metropolitan Engineering Competition” henceforth referred to as MEC, is the annual ESSCO activity comprising an undergraduate engineering competition outlined in this document.

1.1.3 “Pre-internal Qualifiers(s)” refers to the competitions held by ESSCO member schools which are eligible to send competitors to the Ontario Engineering Competition (OEC).

1.1.4 “Chair(s)” refers to the person(s) in the role of overall coordinator of the MEC.

1.1.5 “Organizing Committee”, henceforth referred to as the OC, refers to the team who oversees and organizes the competition as a whole.

1.1.6 “OEC Advisory Board”, henceforth referred to as the OEC-AB, refers to the body that provides advice, guidance, and support to the MEC OC and is the final ruling authority on issues that will affect more than one MEC.

1.1.7 “Vice President of Competitions” henceforth referred to as VP Competitions, or equivalent position, refers to the role that oversees all the Competition Coordinator(s) and all overarching responsibilities related to the competitions.

1.1.8 “Competition Coordinator(s)” refers to the role responsible for the specific competition within a competition section of the rulebook. The Competition Coordinator(s) are appointed or chosen by a process set forth by the Chair(s). The Competition Coordinator(s) are the primary means of contact for competitors, judges, and a liaison between the OC.

1.1.9 “Rules Violation Appeals Committee” is a committee to which competitors can appeal a decision regarding (a) rule violation(s) and is defined in Section 1.8 (Violation of Competition Rules).

1.1.10 “Competition Team”, henceforth referred to as “team”, refers to the team competing at MEC, as qualified through an Internal Qualifier.

1.1.11 “Competitor” refers to an individual on a Competition Team competing at MEC, as qualified through an Internal Qualifier.

1.1.12 “Competition Package” refers to the specific information that Competitors are given for their respective competition.

1.1.13 “Canadian Engineering Competition” henceforth referred to as CEC, the competition which the winning teams at OEC are invited to.

1.2 Eligibility

1.2.1 The eligibility of all competitors, teams, and projects is determined in accordance with the following guidelines set out in this rulebook.

1.2.2 The Chair(s) may grant exceptions to the guidelines set out in this rulebook, while honoring the spirit of these guidelines.

1.2.2 Competitor Qualification

1.2.2.1 All Competitors and Competition Teams must have qualified during a Pre-internal Qualifier.

1.2.2.2 The qualifying teams must be composed of members from the same ESSCO member school that hosted the Pre-internal Qualifier.

a. The maximum team size outlined in the competition-specific sections shall be the maximum team size of the OEC internal qualifier.

1.2.2.3 First place teams from a Pre-internal Qualifier are eligible to compete.

1.2.2.4 The team appearing at OEC must comprise the original team as it competed at its Pre-internal Qualifier.

a. Innovative Design and Engineering Communications teams must present the same project as their Pre-internal Qualifiers.

1.2.2.5 If a competitor cannot attend OEC, the team may compete in their absence as a reduced team or may find a suitable replacement. This may only be done in cases of teams with three or more members, and at the discretion of the Chair(s).

1.2.2.6 If a first-place team cannot attend OEC or meet this criteria, the next runner-up from the same Pre-internal Qualifier may attend OEC, at the discretion of the Chair(s).

1.2.2.7 All competitors must be ESSCO members at the time of the competition with the following exceptions:

a. In the cases of Innovative Design and Engineering Communications, competitors may have graduated no more than three (3) months prior to the competition provided the project

entered was completed before graduation.

i. A letter confirming no changes have been made to the project since the competitors graduation is required from the supervising professor. This letter must be dated no more than 30 days prior to the competition and be submitted to the Chair(s) with the competition abstract.
ii. Competitors not presently enrolled in classes or on co-op are still eligible provided they are considered an enrolled/active student by their institution.

1.3 Regulation of Amendments

1.3.1 These general regulations shall be considered the official regulations of MEC. These regulations can only be amended with the approval of the OEC-AB.

1.3.2 The layout and formatting of this document may be altered by the OC without approval of the OEC-AB.

1.4 Violation of Competition Rules

1.4.1 Rule Violation Reporting System

1.4.1.1 In the event of a suspected rule violation the following process shall occur:

a. The Rule Violation Reporting System shall be a form created by the Rule Violation Committee and made accessible to all competitors during the MEC and at least 24 hours afterwards.
b. The Rule Violation form shall include the following:
   i. Name of Reporting Individual;
   ii. Competition/Position of Reporting Individual;
   iii. Alias of Reporting Individual (if applicable);
   iv. Rule Violation Date and Time;
   v. Name of the Individual Whom the Rule Violation Report is Against;
   vi. Competition/Position of Person Rule Violation Report is Against;
   vii. Rule Violation Details (including references to rulebook sections and backup information if applicable);
   viii. Rule Violation Witness(es), if applicable; and
   ix. Contact Information.
c. The competitor(s) shall be notified verbally and in writing immediately upon the receipt of a rule violation submission against them. The competitor(s) shall be allowed to continue the competition and work on their solution.
d. Competitors shall be provided, in writing, the alleged violation of the rules following the design phase.
e. If the competitor(s) are found guilty of a violation following the conclusion of the design phase, competitor(s) have one (1) hour to appeal the Rule Violation Committee's (RVC) decision to the Rule Violation Appeal Committee (RVAC).

i. Appeals must be made within one (1) hour of receiving the RVC's decision, or by 9am the following morning if the decision by the RVC was made after 12am.

f. In the event of the discovery of a rule violation following the competition, reports should be made via email to the Chair(s) following the requirements of the form above. Competitor(s) will be notified upon receipt of the rule violation submission. The RVC shall provide a report with their findings to the competitor(s) within 48 hours. Appeals to these post competition submissions must be made to the RVAC within 48 hours.

1.4.2 Rule Violation Committee

1.4.2.1 An RVC shall be formed for each year's MEC and approved by the OEC-AB no later than the last OEC-AB meeting prior to the competition weekend.

1.4.2.2 Committee Membership

a. The RVC shall consist of four (4) members including:
   i. One member of the OEC OC who shall serve as the RVC Chair;
      - The RVC Chair shall be a non-voting member of the RVC who leads discussions and handles all official communications.
   ii. Three members appointed by the OEC Chair(s) or VP Competition(s).

1.4.2.3 Committee Response Procedure

a. To ensure that the RVC has a complete understanding of the Rule Violation, they should review the Rule Violation submission in full and ask any follow up questions as required to:
   i. Name of the Individual Whom the Rule Violation Submission is Against;
   ii. The Reporting Individual;
   iii. Any witnesses to the Rule Violation, if applicable/appropriate; and
   iv. The Competition Coordinator(s) if applicable/appropriate.
b. The RVC shall use their best judgement, in consultation with this rulebook, to determine the appropriate response/penalties. All RVC decisions require a majority vote.
c. Any official communications shall be made by the Chair of the RVC, with the knowledge and approval of all members of the RVC. Response to any official communication shall be shared with all the RVC members.

1.4.2.4 Conflict of Interest

a. A conflict of interest shall be defined as a strong personal relationship between a member of the RVC and the Individual Reporting, any individual(s) affected by the rule violation submission, or individuals involved in the rule violation that unfairly biased their actions or decisions, or a situation in which a person is in a position to derive personal benefit from actions or decisions made in their official capacity. A conflict of interest includes but it not limited to:

i. A vetted interest in the outcome of the RVC proceedings;
ii. Any sort of romantic or friendly relationship;
iii. Personal conflicts in the past or present.

b. A member of the RVC should remove themselves from the RVC if they believe they have a perceived or real conflict of interest. Should anyone believe that an RVC member has a conflict of interest, it should be presented to the RVC. If one member of the RVC believes that a perceived or real conflict of interest exists, the appropriate member shall be removed from the RVC.

c. The remaining members of the RVC will be responsible for choosing a replacement member on a temporary or permanent basis, as appropriate, and notifying the Appeal Board of the change. The remaining members of the RVC may also decide to release changes of membership to the MEC competitors should they deem it appropriate.

1.4.3 Appeal of Decision

1.4.3.1 An RVAC shall be formed for each year's OEC and approved by the OEC-AB no later than the last OEC-AB meeting prior to the competition weekend.

1.4.3.2 Committee Membership

a. The RVAC shall consist of four (4) members including:
   i. One member of the OEC OC who shall serve as the RVC Chair;
      - The RVAC Chair shall be a non-voting member of the RVAC who leads discussions and handles all official communications.
   ii. One member of ESSCO BoD appointed by the OEC Chair(s);
   iii. Two members of the OEC OC appointed by the OEC Chair(s) or VP Competition(s).

1.4.3.3 Committee Membership

a. The RVAC shall hear all appeals resulting from decisions made by the RVC. The Chair of the RVC should attend the meeting where an appeal is heard to provide information, but should not participate in any decision made regarding the appeal.
b. The RVAC may uphold, overturn or expand upon decisions made by the RVC, and are expected to take the recommendations of the RVC into consideration when making their decision. All decisions made by the RVAC are final.

1.4.3.4 Conflict of Interest

a. A member of the RVAC should remove themselves from the Appeal Process if they believe they have a conflict of interest as defined above in point 1.4.2.4 Conflict of Interest. The same process as in 1.4.2.4 shall be followed.

1.4.4 Recording of Information

1.4.4.1 The following information must be kept by the OEC OC after the RVC and RVAC have come to a decision. This information shall be retained indefinitely.

a. All correspondence, notes, and other information taken during the investigation;
b. The decision that was reached and why; and
c. The recommendations to the RVAC.

1.4.4.2 All information recorded shall be stored on the OEC Chair(s) drive, with access given to the OEC-AB Chair.

1.5 School Anonymity

1.5.1 School anonymity is in place to avoid any prejudice or favoritism towards a given team by the judges.

1.5.2 Competitors are also forbidden from wearing any identifying symbols of their respective schools.

1.5.3 Each team shall use the alias assigned by their Competition Coordinator(s).

1.5.4 No information in a team's presentation may refer to the team's school.

1.5.5 When submitting deliverables, the content and name must not disclose the name of the team's school.

1.6 Language

1.6.1 The official language of the OEC is English.

1.6.2 French is recognized as a secondary language that will be accommodated upon request.

a. At the time of registration, competitors must notify the OC that they choose to compete in French.
b. The OC will provide all necessary documentation in French and provide live translation to make the competition fair and feasible for the competitor.

1.6.3 All deliverables must be consistent in language.

1.6.4 All members of a team must compete in the same language.

1.7 Penalties

1.7.1 If the identity of a team's school is disclosed in the presentation slides or name, the presenting team will be penalized with a loss of ten (10) points.

1.7.2 If the team reveals their school identity to the judges before the conclusion of the presentation phase, the first offense will result in a ten (10)-point penalization, and the second will result in disqualification.

1.7.3 Delegates and audience members associated with competing schools are forbidden from revealing - explicitly or implicitly - a school's identity in any of the presentation rooms at any time.

a. This ban includes but is not limited to apparel, signage or any other material that bears the name, crests, colors or other identifying symbols of a competing university. The team associated with an individual who breaks this rule is subject to a loss of ten (10) points.
b. In the case of Parliamentary Debate, revealing any team's identity or respective school will result in the revealing team's loss of the current round, at the Moderator's discretion.

1.7.4 In the case of Parliamentary Debate, revealing any team's identity or respective school will result in the revealing team's loss of the current round, at the Moderator's discretion.

1.7.5 Teams may not be in the audience for presentations of their competition until after they have completed their presentation. Any attempts at entering the presentation room before their allotted time will result in a warning for a first offense, and a ten (10) point penalty for any further offenses.

1.7.6 In the case of Junior Design and Senior Design, given that the intention of the competition is to create a practicable, realistic solution to the given problem statement, a penalty of fifty (50) points will be deducted from the final score of any team whose design is based fundamentally upon an impossible concept, and/or makes no realistic attempt to solve the design problem within the constraints of the problem statement.

1.7.7 If a team has been found guilty of plagiarism, the team shall be disqualified from the competition, and their home school shall be notified within one (1) hour.

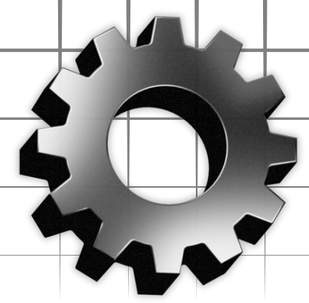
1.7.8 Teams will not be allowed to extend past their allotted presentation time. Presentations will be cut-off when the time limit has been reached.

1.7.9 For every five minutes late on submission of a competition deliverable, the team will be subject to a ten (10) point deduction.

2. GENERAL COMPETITION STANDARDS

2.1 Competition Coordinators

2.1.1 The Competition Coordinator(s) are responsible for the design and implementation of their respective competition.



2.1.2 The Competition Coordinator(s) must be present at all of the presentations.

2.1.3 The Competition Coordinator(s) must present the design problem at the beginning of the competition and answer any questions raised by competitors.

2.1.4 The Competition Coordinator(s) will be available to competitors and judges during competition hours for questions and requests.

2.2 Official Timekeeper

2.2.1 The Official Timekeeper must be responsible for enforcing time limits during the question period, solution development, and presentations.

2.2.2 This role can be held by the Competition Coordinator(s)

2.2.3 The responsibilities of the Parliamentary Debate Timekeeper are outlined in Section 11.1.3.

2.3 Judges

2.3.1 All competitions must have a minimum of one (1) judge.

2.3.2 Judges are required to assess the problem-solving abilities, proposed solution, and communication skills of the competitors.

2.3.3 Judges in Communications should be comprised of non-technical backgrounds including sales, consulting, and communications.

2.3.4 Judges in Parliamentary Debate and Junior Design should come from a variety of backgrounds capturing technical and non-technical experience related to the topic.

2.4 Competition Package

2.4.1 Each competitor will be given the Competition Package prior to the competition that will contain any specific information regarding the competition.

2.4.1.1 The Competition Package may be compiled into the Delegate Package.

2.4.1.2 The Competition Package should include the following components:

a. Competition Abstract
b. Competition Rules;
c. Competition Rubric;
d. Competition Logistics including; and

i. Room locations;
ii. Personnel; and
iii. Awards.

e. Competition Procedure.

2.4.2 It is the competitor's responsibility to ensure that they have received the package.

2.4.3 At the competition briefing, each team will receive a package outlining the problem definition, design and presentation requirements, rules, marking scheme, and any other information deemed necessary by the Competition Coordinator(s).

2.4.3.1 Competitions with pre-weekend deliverables will receive a reduced version of this package at the start of their work period.

2.5 Integrity of the Competition

2.5.1 Since the use of the Internet and other external resources is permitted in this competition, all information used by competitors must be referenced using a consistent and conventional method.

2.5.2 Competitors are not permitted to submit work completed by anyone other than the members of their team.

2.5.2.1 In the case of Engineering Communications and Innovative Design, an authorization letter must be provided as discussed in the subsequent sections of this rulebook.

2.5.3 If there is any evidence that competitors are submitting plagiarized work, the entire team will be eliminated from the competition.

2.5.4 Competition Coordinator(s) will monitor each team during the design process to deter teams from cheating.

2.5.5 Rules for the use of Generative AI will be outlined in the Competition Packages.

2.6 Resource Requests

2.6.1 Prior to the competition, the team may submit a resource request if they require resources outside of those listed in the subsequent competition sections. All resource requests must be made no later than twenty-four (24) hours before the competition. The request shall be formatted with 12-point

font, 1.5 line spacing, 1” margins, Serif font(s), and is limited to one (1) page. The request shall include at a minimum:

a. Team Name
b. Team Members
c. School
d. Resource being requested
e. Reason for requested resource

2.6.2 Requests will be reviewed by the VP Competition(s) to determine if the request is viable. The team will be informed on whether their request will be accommodated at a minimum twelve (12) hours before the competition.

2.7 Timeline

2.7.1 Pre-Competition

2.7.1.1 The following will be provided to Competitors prior to the competition:

a. The Competition Package (also to be provided to the judges);
b. Day 1 and Day 2 schedules; and
c. Exact specifications of equipment available to teams during the competition.

2.7.1 Competition Briefing

2.7.2.1 The problem must be presented to all competitors at the beginning of the competition. The Competition Coordinator(s) must provide detailed explanations of what is expected from the competitors, both orally and in writing. The presentation of the problem will take no longer than fifteen (15) minutes.

2.7.2.2 After the conclusion of the presentation, there will be a period of fifteen (15) minutes for competitors to ask the Competition Coordinator(s) any questions.

2.7.2.3 The answers will be provided orally and recorded in written form.

2.7.2.4 The Competition Briefing shall be scheduled for no less than thirty (30) minutes.

2.8 Competition Duration

2.8.1 The competitions shall have a set amount of time to develop their solutions, produce all required deliverables, and prepare their presentations.

2.8.2 All deliverables shall be submitted to the Competition Coordinator before the end of the allotted time. Competitors may finish before the end of the allotted time.

2.8.3 Competitors are allowed to ask any questions during the competition phase, but it is at the discretion of the Competition Coordinator(s) whether they choose to answer them or not.

2.8.3.1 If there are multiple days of activities for a competition, competitors shall be given a minimum 8 hours of a break between the end of mandatory scheduled competition activity and the beginning of mandatory activities the following day.

2.8.4 All team members must be present and participate in the presentation.

2.8.5 In order to ensure that all competitors cease to work on the case solutions once the design time has ended, the competitors cannot include any material in their oral presentation which is not included within their submitted written reports or presentation materials.

2.9 Timekeeping

2.9.1 Design

2.9.1.1 The design period will begin immediately upon the conclusion of the question period.

2.9.1.2 The remaining time must be periodically announced by the Competition Coordinator(s) before the end of the allotted design time.

2.9.2 Presentation

2.9.2.1 Time is halted when a judge asks a question and when a team member answers the question during the presentation.

2.9.2.2 The remaining time must be indicated to the competitors periodically before the end of the allotted presentation time.

2.9.2.3 A visual countdown may be given to the presenters during the allotted presentation time.

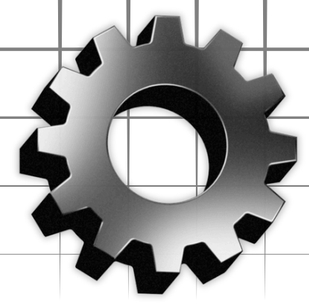
2.9.2.4 After the allotted presentation time has expired, competitors will have a fifteen (15) second grace period to conclude their presentation before they are cut-off to begin the question period.

2.9.2.5 Teams who are cut-off will be penalized.

2.9.2.6 Time to both ask and answer questions will be counted during the question period.

2.10 Presentation Order

2.10.1 Presentation order shall be determined randomly.



2.10.2 Presentation order shall be announced at check-in on Day 2. All teams are required to be present at this announcement.

2.10.3 Teams are not allowed to switch places in the presentation order.

2.10.4 Teams may not be in the audience for presentations.

2.11 Response to Questions

2.11.1 The Competition Coordinator(s) and VP Competition(s) may answer questions during the problem presentation and design phases.

2.11.2 During the presentation of the problem, the Competition Coordinator(s) shall answer orally.

2.11.3 During the design phase, only questions related to deliverable content shall be answered. No answers shall be provided in response to questions about the problem that might lead to the development of a new approach, or which might invalidate a solution.

2.12 Assessment

2.12.1 Judging feedback shall be provided to each Competing Team following the conclusion of MEC.

3. BIO-ENGINEERING

3.1 Overview

3.1.1 In this competition, teams comprehensively solve a bio-engineering problem. Teams are required to identify the issues outlined in the prompt and develop a bio-engineering design process in order to solve the issue in question.

3.1.2 The maximum number of individuals for a Bio-Engineering team is two (2).

3.2 Prompt

3.2.1 It is recommended that the topic incorporate more than one engineering discipline. However, while the topic will challenge competitors' technical knowledge and skills, it will also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer.

3.2.2 Competition topics will be ones that could exist in the real world. Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

3.3 Facilities and Equipment

3.3.1 The following resources are required to be provided for Bio-Engineering teams during the competition phase:

a. One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team;
b. One (1) table per team;
c. One (1) chair per Competitor;
d. One (1) of a whiteboard, chart paper, or blackboard;
e. Paper and pencils/pens for writing (upon request);
f. Internet connectivity; and
g. Power source for computers.

3.3.2 The following resources are required to be provided for Bio-Engineering teams during the presentation phase:

a. One (1) room;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector; and
d. Whiteboard(s) and/or blackboard(s) and associated writing implements.

3.4 External Resources

3.4.1 The competitors are allowed to bring the following equipment with them during the Competition phase:

a. Any background research conducted by team members prior to the competition;
b. Any textbooks, course notes, or other reference material; and
c. Each team member is allowed one device*.

* Devices may include a computer, iPad, cell phone, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. For example, a competitor may use a laptop for research and switch to using an iPad to sketch their design, etc.

3.4.2 Teams are permitted to bring a slideshow template for use in the competition.

3.4.2.1 Templates may contain background images and a basic slide layout.

3.4.2.2 Slides are to have no specific content pre-competition.

3.5 Competition & Deliverables

3.5.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across:

3.5.2 Durations

3.5.2.1 Competition Duration: eight (8) hours

3.5.2.2 Presentation Duration: ten (10) minutes

3.5.2.3 Question Period Duration: five (5) minutes

3.5.3 Deliverables

3.5.3.1 The deliverable requirements will be outlined in the Competition Package. They will include:

a. Report of less than ten (10) pages (excluding appendices); and
b. Presentation.

3.6 Solution & Rubric

3.6.1 The solution should have real-world applicability and forethought.

Solution	• Addresses Problem Statement • Environmental, Biomedical and Economic Consideration • Technical Feasibility • Innovation • Real-world Applicability	/60
Report	• Clarity • Writing Style & Professionalism • Design Justification	/15
Presentation	• Voice • Articulation and Timing • Visual Aids • Response to Questions	/25
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

3.6.2 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

4. CONSULTING ENGINEERING

4.1 Overview

4.1.1 The Consulting Engineering competition will challenge competitors to design a detailed solution to a large-scale engineering problem. The judges represent a mock client, to whom the solution will be proposed.

4.1.2 The maximum number of individuals for a Consulting Engineering team is four (4).

4.2 Prompt

4.2.1 The topic will incorporate more than one engineering discipline. The topic will challenge competitors' technical knowledge and skills, it should also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer.

4.2.2 Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

4.3 Facilities and Equipment

4.3.1 The following resources are required to be provided for Consulting Engineering teams during the competition phase:

a. One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team;
b. One (1) table per team;
c. One (1) chair per Competitor;
d. One (1) of a whiteboard, chart paper, or blackboard;
e. Paper and pencils/pens for writing (upon request);
f. Internet connectivity; and
g. Power source for computers.

4.3.2 The following resources are required to be provided for Bio-Engineering teams during the presentation phase:

a. One (1) room;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector; and
d. Whiteboard(s) and/or blackboard(s) and associated writing implements.

4.4 External Resources

4.4.1 The competitors are allowed to bring the following equipment with them during the Competition phase:

a. Any background research conducted by team members prior to the competition;
b. Any textbooks, course notes, or other reference material; and
c. Each team member is allowed one device*.

* Devices may include a computer, iPad, cell phone, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. For example, a competitor may use a laptop for research and switch to using an iPad to sketch their design, etc.

4.4.2 Teams are permitted to bring a slideshow template for use in the competition.

4.4.2.1 Templates may contain background images and a basic slide layout.

4.4.2.2 Slides are to have no specific content pre-competition.

4.5 Competition & Deliverables

4.5.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across:

4.5.2 Durations

4.5.2.1 Competition Duration: eight (8) hours

4.5.2.2 Presentation Duration: ten (10) minutes

4.5.2.3 Question Period Duration: five (5) minutes

4.5.3 Deliverables

4.5.3.1 The deliverable requirements will be outlined in the Competition Package. They will include:

a. Report of less than ten (10) pages (excluding appendices); and
b. Presentation.

4.6 Solution & Rubric

4.6.1 The winning solution will not necessarily be the most technically effective solution, but the solution that has the most real-world applicability and forethought.

Solution	• Addresses Problem Statement • Environmental, Economic, and Social Analysis • Design Feasibility • Risk Analysis	/60
Report	• Clarity • Writing Style & Professionalism • Deliverable Compliance with Expectations	/15
Presentation	• Voice • Articulation and Timing • Visual Aids • Response to Questions	/25
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

4.6.2 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

5. RE-ENGINEERING

5.1 Overview

5.1.1 Re-engineering is the act of taking an existing engineering concept, product, technique, or technology and incrementally improving on its design to suit an alternate situation or application. In this competition, students will be required to apply the re- engineering process to an existing gadget or mechanism in order to add new functionality to it or enhance its original functionality. The intent of these improvements will be to extend the use of the device to an alternative situation or use case.

5.1.2 The maximum number of individuals for a Re-Engineering team is two (2).

5.2 Prompt

5.2.1 It is recommended that the topic incorporate more than one engineering discipline. While the topic should challenge competitors' technical knowledge and skills, it should also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer.

5.2.2 Competition topics will be ones that could exist in the real world. Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

5.2.3 Cases

5.2.1. The competition will consist of two (2) sets of written cases.

5.2.2. The first case set will be distributed to the competitors seven (7) days in advance of the competition weekend opening ceremonies.

5.2.3. The second case set will be presented to the competitors during the Competition Briefing

5.2.4. Both cases are due at the same time (the end of the competition phase).

5.2.5 The competition deliverables (report and presentation) must include both cases.

5.2.5. During the week prior to the competition start time, all questions posed to the MEC OC regarding the first case must be answered by the MEC OC within twenty- four (24) hours of submission.

5.3 Facilities and Equipment

5.3.1 The following resources are required to be provided for Re-Engineering teams during the competition phase:

a. One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team;
b. One (1) table per team;
c. One (1) chair per Competitor;
d. One (1) of a whiteboard, chart paper, or blackboard;
e. Paper and pencils/pens for writing (upon request);
f. Internet connectivity; and
g. Power source for computers.

5.3.2 The following resources are required to be provided for Re-Engineering teams during the presentation phase:

a. One (1) room;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector; and
d. Whiteboard(s) and/or blackboard(s) and associated writing implements.

5.4 External Resources

5.4.1 The competitors are allowed to bring the following equipment with them during the Competition phase:

a. Any background research conducted by team members prior to the competition;
b. Any textbooks, course notes, or other reference material; and
c. Each team member is allowed one device*.

* Devices may include a computer, iPad, cell phone, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. For example, a competitor may use a laptop for research and switch to using an iPad to sketch their design, etc.

5.4.2 Teams are permitted to bring a slideshow template for use in the competition.

5.4.2.1 Templates may contain background images and a basic slide layout.

5.4.2.2 Slides are to have no specific content pre-competition.

5.5 Competition & Deliverables

5.5.1 MEC. Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across:

5.5.2 Durations

5.5.2.1 Competition Duration: eight (8) hours

5.5.2.2 Presentation Duration: fifteen (15) minutes

5.5.2.3 Question Period Duration: seven (7) minutes

5.5.3 Deliverables

5.5.3.1 The deliverable requirements will be outlined in the Competition Package. They will include:

a. Report of less than one (1) page per case (excluding appendices); and
   i. Competitors may elect to shorten one of the reports and use the remaining space for their other report, provided both reports are written end-to-end in a single document and the document does not exceed two (2) single-sided pages.
b. Presentation

5.6 Solution & Rubric

5.6.1 The winning solution will not necessarily be the most technically effective solution, but the solution that has the most real-world applicability and forethought.

Solution	• Addresses Problem Statement • Environmental, Economic, and Social Analysis • Design Feasibility • Risk Analysis	CASE 1 /40 CASE 2 /20
Report	• Clarity • Writing Style & Professionalism • Deliverable Compliance with Expectations	/15
Presentation	• Voice • Articulation and Timing • Visual Aids • Response to Questions	/25
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

4.6.2 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

6. PROGRAMMING

6.1 Overview

6.1.1 The goal of the programming category is to encourage engineering students to produce a piece of readable software. The teams will use their software development skills, their technical writing abilities, and their project management skills to design a solution to a posed problem. The judges will represent mock company executives for project approval.

6.1.2 The maximum number of individuals for a Programming team is four (4).

6.2 Prompt

6.2.1 It is recommended that the topic incorporate more than one engineering discipline. While the topic should challenge competitors' technical knowledge and skills, it should also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer.

6.2.2 Competition topics will be ones that could exist in the real world. Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

6.3 Facilities and Equipment

6.3.1 The following resources are required to be provided for Programming teams during the competition phase:

a. One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team;
b. One (1) table per team;
c. One (1) chair per Competitor;
d. One (1) of a whiteboard, chart paper, or blackboard;
e. Paper and pencils/pens for writing (upon request);
f. Internet connectivity; and
g. Power source for computers.

6.3.2 The following resources are required to be provided for Programming teams during the presentation phase:

a. One (1) room;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector; and
d. Whiteboard(s) and/or blackboard(s) and associated writing implements.

6.4 External Resources

6.4.1 The competitors are allowed to bring the following equipment with them during the Competition phase:

a. Any background research conducted by team members prior to the competition;
b. Any textbooks, course notes, or other reference material; and
c. Each team member is allowed one device*.

* Devices may include a computer, iPad, cell phone, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. For example, a competitor may use a laptop for research and switch to using an iPad to sketch their design, etc.

6.4.2 Teams are permitted to bring a slideshow template for use in the competition.

6.4.2.1 Templates may contain background images and a basic slide layout.

6.4.2.2 Slides are to have no specific content pre-competition.

6.4.3 Software Resources

6.4.3.1 Teams may not use pre-existing code they copied from the internet to solve any portion of the competition.

6.4.3.2 The judges hold the right to ask any team member to describe what a particular section of the code does at any given point during the presentation.

6.5 Competition & Deliverables

6.5.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across:

6.5.2 Durations

6.5.2.1 Competition Duration: eight (8) hours

6.5.2.2 Presentation Duration: ten (10) minutes

6.5.2.3 Question Period Duration: five (5) minutes

6.5.3 Deliverables

6.5.3.1 The deliverable requirements will be outlined in the Competition Package. They will include:

a. Code, used to evaluate the team's solution; and
b. Presentation.

6.6 Solution & Rubric

6.6.1 The winning solution will not necessarily be the most technically correct solution, but the solution that has the most real-world applicability and forethought.

6.6.2 The solution/code will be tested IAW the Competitors Package.

Solution	• Personalization & Adaptability • Deliverable Compliance with Expectations • User Experience • Innovation and Creativity • Readability	/40
Report	• Efficiency • Structure • Testing • Design Process and Justification	/35
Presentation	• Voice • Articulation and Timing • Visual Aids • Response to Questions	/25
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

6.6.3 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

7. JUNIOR DESIGN

7.1 Overview

7.1.1 This competition challenges junior engineering students to design, build, and test a prototype to address a technical problem. The Junior Team Design category is similar to the Senior Team Design competition, but emphasis is placed on prototype functionality rather than design theory.

7.1.2 The maximum number of individuals for a Junior Design team is four (4).

7.1.3 Junior Design teams must be entirely composed of students who have not yet completed 50% or more of the minimum number of credits required to complete their respective accredited engineering programs.

7.2 Facilities and Equipment

7.2.1 The following resources are required to be provided for Junior Design teams during the competition phase:

a. One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team;
b. One (1) table per team;
c. One (1) chair per Competitor;
d. One (1) of a whiteboard, chart paper, or blackboard;
e. Paper and pencils/pens for writing (upon request);
f. Internet connectivity; and
g. Power source for computers.

7.2.2 The following resources are required to be provided for Junior Design teams during the presentation phase:

a. One (1) room;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector;
d. Whiteboard(s) and/or blackboard(s) and associated writing implements; and
e. One (1) table, for displaying prototype.

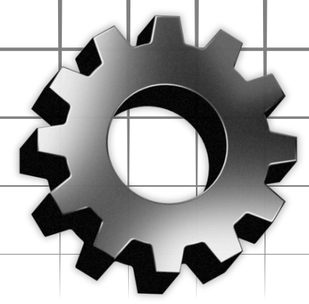
7.3 External Resources

7.3.1 The competitors are allowed to bring the following equipment with them during the Competition phase:

a. Any background research conducted by team members prior to the competition;
b. Any textbooks, course notes, or other reference material; and
c. Each team member is allowed one device*.
d. Camera

* Devices may include a computer, iPad, cell phone, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. For example, a competitor may use a laptop for research and switch to using an iPad to sketch their design, etc.

7.3.2 Teams are permitted to bring a slideshow template for use in the competition.



7.3.2.1 Templates may contain background images and a basic slide layout.

7.3.2.2 Slides are to have no specific content pre-competition.

7.3.3 No external tools can be brought by competitors. They are only permitted to use the external tools provided at the MEC workstations.

7.4 Competition & Deliverables

7.4.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across:

7.4.2 Durations

7.4.2.1 Competition Duration: six (6) hours

7.4.2.2 Presentation Duration: ten (10) minutes

7.4.2.3 Question Period Duration: five (5) minutes

7.4.3 Deliverables

7.4.3.1 The deliverable requirements will be outlined in the Competition Package. They will include:

a. Prototype of design; and
b. Presentation.

7.5 Solution & Rubric

7.5.1 The winning solution will not necessarily be the most technically effective solution, but the solution that has the most real-world applicability and forethought.

7.5.2 Given that the intention of the competition is to create a practicable, realistic solution to the given problem statement, a penalty of fifty (50) points will be deducted from the final score of any

team whose design is based fundamentally upon an impossible concept, and/or makes no realistic attempt to solve the design problem within the constraints of the problem statement.

Design Quality	• Creativity • Scalability • Feasibility • Versatility	/40
Testing		/30
Presentation	• Design Process • Design Justification • Voice • Articulation and Timing • Response to Questions	/30
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

7.5.3 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

8. SENIOR DESIGN

8.1 Overview

8.1.1 Students are given a complex engineering problem and are required to design and build a working prototype. The teams will then present their solution and test their prototypes in front of a panel of judges.

8.1.2 The maximum number of individuals for a Senior Design team is four (4).

8.2 Facilities and Equipment

8.2.1 The following resources are required to be provided for Senior Design teams during the competition phase:

a. One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team;

b. One (1) table per team;
c. One (1) chair per Competitor;
d. One (1) of a whiteboard, chart paper, or blackboard;
e. Paper and pencils/pens for writing (upon request);
f. Internet connectivity; and
g. Power source for computers.
h. Materials and tools specific to the design problem; and
i. One (1) competition testing space.

8.2.2 The following resources are required to be provided for Senior Design teams during the presentation phase:

a. One (1) room;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector;
d. Whiteboard(s) and/or blackboard(s) and associated writing implements; and
e. One (1) table, for displaying prototype.

8.3 External Resources

8.3.1 The competitors are allowed to bring the following equipment with them during the Competition phase:

a. Any background research conducted by team members prior to the competition;
b. Any textbooks, course notes, or other reference material; and
c. Each team member is allowed one device*.
d. Camera

* Devices may include a computer, iPad, cell phone, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. For example, a competitor may use a laptop for research and switch to using an iPad to sketch their design, etc.

8.3.2 Teams are permitted to bring a slideshow template for use in the competition.

8.3.2.1 Templates may contain background images and a basic slide layout.

8.3.2.2 Slides are to have no specific content pre-competition.

8.3.3 No external tools can be brought by competitors. They are only permitted to use the external tools provided at the MEC workstations.

8.4 Competition & Deliverables

8.4.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across:

8.4.2 Durations

8.4.2.1 Competition Duration: ten (10) hours

8.4.2.2 Presentation Duration: ten (10) minutes

8.4.2.3 Question Period Duration: five (5) minutes

8.4.3 Deliverables

8.4.3.1 The deliverable requirements will be outlined in the Competition Package. They will include:

a. Prototype of design; and
b. Presentation.

8.5 Solution & Rubric

8.5.1 The winning solution will not necessarily be the most technically effective solution, but the solution that has the most real-world applicability and forethought.

8.5.2 Given that the intention of the competition is to create a practicable, realistic solution to the given problem statement, a penalty of fifty (50) points will be deducted from the final score of any

team whose design is based fundamentally upon an impossible concept, and/or makes no realistic attempt to solve the design problem within the constraints of the problem statement.

Design Quality	• Aesthetics • Creativity & Innovation • Code • Overall Design	/40
Testing		/30
Presentation	• Design Process • Design Justification • Voice • Articulation and Timing • Response to Questions	/30
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

8.5.3 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

9. ENGINEERING COMMUNICATIONS

9.1 Overview

9.1.1 The goal of this competition is to describe an engineering topic (i.e. process, product, device, issue) in terms that the general public can understand. The Engineering Communications competition is prepared entirely outside the realm of MEC. Competitors research and prepare their own presentation topic.

9.1.2 The maximum number of individuals for an Engineering Communications team is two (2).

9.2 Topic & Project Eligibility

9.2.1 The presentation topic should fulfill the following criteria:

a. The topic must be technical in nature; and

b. The presentation will assess social, environmental, political, and economic impacts of the topic.

9.3 Deliverables

9.3.1 The team must submit a presentation abstract prior to MEC. The purpose of this abstract is to brief judges and organizers on the topic the team will be presenting. The abstract must:

a. Be a maximum of two hundred (200) words;

i. As there is a word cap, formatting is at the discretion of the team.

b. The presentation will assess social, environmental, political, and economic impacts of the topic.
c. State the competitor's name(s); and
d. Include the team's alias.

9.3.2 Due dates will be assigned at the discretion of the Competition Coordinator(s) to allow time for testing the presentation and printing hard copies. The hard copies of slides will be used to aid the judges in keeping notes and providing feedback when necessary and desired.

9.4 Facilities & Equipment

9.4.1 There is no competition phase for Engineering Communications. Competitors will have deliverables due prior to MEC. During MEC the competitors will participate in the presentation phase.

9.4.2 The following resources are required to be provided for Engineering Communication teams during the presentation phase:

a. One (1) large workroom;
b. One (1) computer containing the team's presentation file;
c. One (1) digital projector; and
d. Whiteboard(s), and/or blackboard(s), and associated writing implements.

9.4.3 If the team requires additional resources to those provided at any point during MEC, competitors should refer to the Resource Request section.

9.5 External Resources

9.5.1 Competitors are required to bring any presentation or display materials which they will use, including pictures or diagrams.

9.6 Competition Durations

9.6.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across MEC.

9.6.2 Presentation duration: twenty (20) minutes.

9.6.3 Question Period Duration: five (5) minutes.

9.7 Solution & Rubric

Introduction to Topic	• Topic and Background Clearly Presented • Appropriate Level of Language • Interest Elicited by Topic • Accuracy of Explanation	/25
Critical Analysis	• Environmental Analysis • Social Analysis • Economic Analysis	/30
Written Abstract	• Concise and Captivating Introduction of Topic	/5
Presentation	• Voice • Articulation and Timing • Visual Aids • Response to Questions	/25
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

9.7.3 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

9.7.2 Additional Penalties

9.7.2.1 For each grammar or spelling mistake in the abstract or presentation, one (1) point deduction.

9.7.2.2 For each day of late submission for the abstract, five (5) points deduction.

9.7.2.3 For presentations less than seventeen (17) minutes, five (5) points/minute deduction for every minute under seventeen (17) minutes (e.g. a fifteen (15) minute presentation will result in a deduction of ten (10) points.

10. INNOVATIVE DESIGN

10.1 Overview

10.1.1 The innovative design competition is prepared entirely outside the realm of the MEC. Competitors choose their own topic, prepare research and develop a design. The designs must be new and innovative and address a void in society.

10.1.2 The maximum number of individuals for an Innovative Design team is six (6).

a. This maximum includes competitors both present at MEC and those who have signed the letter of authorization.

10.2 Topic & Project Eligibility

10.2.1 The presentation topic should fulfill the following criteria:

a. The topic must be technical in nature;
b. The project must have a design component; and
c. The presentation will assess social, environmental, political, and economic ramifications of implementation.

10.2.1.1 While the above list provides a guideline to project selection, no projects will be disallowed due to the topic selection. The judging criteria provides a component which analyzes the applicability of the project's topic.

10.2.2 Competitors must present the same topic in the OEC.

10.2.3 Work done during employment related to the engineering degree is eligible provided that an authorization letter is obtained from the employer allowing the work to be presented at the competition. This letter must be submitted with the competition abstract if applicable.

10.2.4 Should the presenting team be a subset of a project team, the remaining members of the project team must sign a letter authorizing presentation of the project in order for it to be eligible for the competition. This letter must be submitted with the competition abstract if applicable.

10.3 Deliverables

10.3.1 Design Summary

10.3.1.1 The team must submit a design summary prior to MEC. The purpose of this abstract is to brief judges and organizers on the topic the team will be presenting. The design summary must:

a. Be limited to one (1) page in length:
b. Follow conventional formatting characteristics;
    i. 12- point font;

ii. 1.5 line spacing; and
iii. 1′ page margins

c. Include the design topics and descriptions;
d. State the competitor’s name(s); and
e. Include the team’s alias.

10.4 Facilities and Equipment

10.4.1 There is no competition phase for Innovative Design. Competitors will have deliverables due prior to MEC. During MEC, the competitors will participate in the presentation phase.

10.4.2 The following resources are required to be provided for Innovative Design teams during the presentation phase:

a. One (1) workroom;
b. One (1) computer containing the team’s presentation file;
c. One (1) digital projector; and
d. Whiteboard(s), and/or blackboard(s), and associated writing implements.

10.4.3 If the team requires additional resources to those provided at any point during MEC, competitors should refer to the Resource Request section.

10.5 External Resources

10.5.1 Competitors are required to bring any presentation or display materials which they will use, including pictures or diagrams and models or prototypes.

10.6 Competition Durations

10.6.1 Competitors shall refer to section 2.8 Competition Deliverables for general guidelines across MEC.

10.6.2 Presentation Duration: fifteen (15) minutes.

10.6.3 Question Period Duration: five (5) minutes.

10.7 Rubric

Solution	• Design Justification • Environmental, Social, and Economic Feasibility • Technical Feasibility/Innovation	/70
Design Summary	• Clarity • Writing Style and Professionalism • Deliverance and Compliance with Expectations	/10
Presentation	• Voice • Articulation and Timing • Visual Aids • Response to Questions	/20
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -50 -25
Total		/100

10.7.1 This rubric is subject to change. It is simplified to maintain confidentiality before the Competition Case is released. The completed rubrics with full point breakdowns will be given to competitors with the case document at the time of the competition.

10.7.2 Additional Penalties

10.7.2.1 For every minute the presentation exceeds the time limit, two (2) points deduction.

10.7.2.2 For each day of late submission for the design summary, five (5) points deduction.

10.7.2.3 For every three (3) grammar or spelling mistakes in the design summary, one (1) point deduction.

11. PARLIAMENTARY DEBATE

11.1 Overview

11.1.1 Parliamentary Debate Competitors must use analytical techniques to present, with minimum preparation, a reasoned point of view of a resolution that has not been disclosed beforehand.

11.1.2 The goal is to assess the competitors' abilities to convey ideas and develop arguments. The purpose is not to assess competitor knowledge of parliamentary procedure and formal debating rules. Therefore, the rules normally used in debates have been modified and relaxed to enable students with no formal debate experience to take part.

11.1.3 There must be two (2) individuals per Parliamentary Debate team.

11.2 Personnel Required

11.2.1 Competition Coordinator(s)

a. It is recommended that the Competition Coordinator(s) have previous experience in debates, preferably at the MEC or internal qualifier.

11.2.2 Debate Moderator

a. The Debate Moderator, otherwise known as a Moderator, is the expert in debate procedure and interpretation. The Moderator acts as an impartial judge and ensures that the judges understand the rules of the debate.
b. The Moderator of the debate ensures the rules of the debate are upheld. They grant the right to speak (introduces the debater) and enforce the time limits. Most importantly, the Moderator makes rulings on the "points of procedure" put forward by the debaters. The decisions of the Moderator are final.

11.2.3 Timekeeper

a. The Timekeeper is responsible for ensuring that competitors stay within their allotted time limits. The timekeeper will inform speakers of specific time intervals.
b. The Timekeeper will use hand signals to indicate the number of minutes a speaker has left to conclude their argument.
    i. The first and last thirty (30) seconds of a speech will be marked by the Timekeeper banging on the table, with the first bang indicating that Points of Information (POIs) are now permitted and the second bang indicating that POIs are no longer permitted.
    ii. The Timekeeper will also give a ten (10) seconds final countdown with their hands.
    iii. If a question is asked in the first or last minute of the speech, the Timekeeper will indicate this to the Moderator so that it can be stopped.
    iv. If a member of the debater team knocks on the table during the final ten (10) seconds, the Timekeeper will permit an extra and final fifteen (15) seconds of grace.

11.3 Facilities and Equipment

11.3.1 The following facilities and equipment shall be provided for the competition:

a. One (1) amphitheater
b. An area for judge's deliberation

c. Areas for debater's preparation
d. One (1) to Two (2) tables
e. One (1) podium with a microphone
f. One (1) display board, projector, chalkboard or whiteboard is required to display the debate topic during the debate
g. Paper and pencils/pens for writing (upon request);

11.3.2 The layout of the room should place the tables and podium at the front of the room. The first row of the room will be reserved for the judges, Moderator, and Timekeeper.

a. During preparation prior to each debate, the Government team will prepare in the room, while the Opposition team will prepare in the hallway located outside the room.

11.3.3 Competitors may bring their own stopwatches or writing pads.

11.3.4 Other external resources may be allowed at the discretion of the Competition Coordinator(s).

11.3.5 Props are prohibited.

11.4 Debating Rules and Elements

11.4.1 Teams

11.4.1.1 The debate is between two teams, each with two members. The government will be given the task of arguing for the motion. The opposition will then proceed to clash with the motion.

11.4.1.2 Before the resolution is revealed, a coin toss is held. The team that wins the toss is the government.

11.4.1.3 Teams are always expected to act responsibly. Inappropriate language or offensiveness towards the other team, the officials, or the audience is not acceptable.

11.4.1.4 Teams are expected to direct all their speeches towards the judge(s). Teams may assume the judge(s) have general engineering knowledge of a first-year undergraduate engineering student. No facts or statistics that are not common knowledge may be cited during the debate.

11.4.2 Resolutions

11.4.2.1 The resolutions will be developed by the Competition Coordinator(s). They will be related to an issue that the average engineering student should have a defensible opinion on without any preparation.

a. The resolutions will not be truisms. Absolute words such as "all," "everyone," and "always" will also be avoided (e.g. "Be it resolved that all engineers are good at math" is not a good resolution, since there are always exceptions).

11.4.2.2 Once both teams are informed of the resolution, they are given five (5) to ten (10) minutes to prepare for the debate.

11.4.2 Time Allotted to Debaters

11.4.3.1 The speaking order and times will be as follows:

a. First speaker of the government: five (5) minutes
b. First speaker of the opposition: five (5) minutes
c. Second speaker of the government: five (5) minutes
d. Second speaker of the opposition: five (5) minutes
e. Rebuttal by the first speaker of the government: two (2) minutes
f. Rebuttal by the first speaker of the opposition: two (2) minutes.

11.4.3.2 If a debater is over the time limit, the Moderator will allow a fifteen (15) second grace period. Upon the exhaustion of speaking time, loud pounding on tables will be deemed appropriate.

11.4.3.3 If there is a discrepancy between the Timekeeper’s stopwatch and any competitor’s stopwatch, the Timekeeper’s stopwatch will take precedence.

11.4.4 Types of Motions

11.4.4.1 The motions that competitors may encounter will be of the following types:

Abbreviation	Meaning
THW	This House Would
THBT	This House Believes That
THS	This House Supports
THO	This House Opposes
THP	This House Prefers
THR	This House Regrets

11.4.5 Role of the Government

1.4.5.1 The government must argue for the resolution presented in the motion. The resolutions must not be squirreled or converted into truisms.

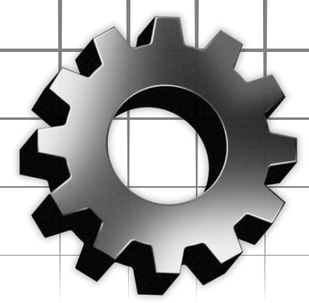
a. Squirreling is the act of redefining the resolution so that it has a meaning different than the one intended by the resolution.

11.4.5.2 The government, if desired, may narrow the scope of the motion and define terms and who the House is, if not already defined in the motion.

11.4.5.3 The government must also avoid specific knowledge debates where the average engineer has no familiarity with the topic. Two examples of different strategies that could be followed in the debate are:

1) The Principle Case (This House Believes That)

a. In a principle case, the government presents a principle and a contention. The principle is a general statement that is debatable based on facts, experience, or morals.
   i. Examples include "Engineers need to be well rounded" and "Canadians abuse the free health-care system."
b. The contention is the application of the principle to a situation relating to the resolution. The contention must be about one major topic and may be a restatement of the resolution.
   i. Example contentions for the above principles would be "Engineering students should be



required to take more complementary studies classes" and "People should be discouraged from going to a doctor unless necessary."

2) The Plan Case (This House Would)

a. In a plan case, the government still identifies a principle and a contention. Once this is done, they present a plan for implementing the contention and changing the status quo. They must identify the need for change and how the plan will induce this change.
    i. Example plans for the above contentions would be “Students should be required to take two years of general studies before entering an engineering program” and “Individuals should be charged a fee every time they visit a doctor.”

b. The plan case is the most effective when the principle and contention are almost non-debatable.
    i. For example, if the resolution is “Be it resolved that waste management should become more stringent,” the contention is almost unquestionable. However, the proposition can intensify the debate by adding, “Thus, we propose that any household that produces more than a certain quota of waste be severely fined.”
c. In a plan case, it is important that the plan does not become too specific since preparation time is limited and the technical knowledge necessary to develop the plan is non-existent.

11.4.6 The First Speaker

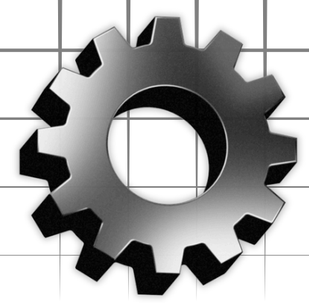
11.4.6.1 The first speaker of the government must explain the interpretation of the resolution, clearly state the principle and contention, and clarify any definitions. If a plan case is introduced, the entire plan must be outlined in the first speaker's speech. Finally, the first speaker must initiate the argumentation for the contention and plan.

11.4.7 The Second Speaker

11.4.7.1 The second speaker of the government continues the argumentation of the first speaker and rearms concepts that have been attacked by the opposition. New parts of a plan may not be introduced and terms in the resolution may not be defined. New lines of argumentation and new evidence, however, may be introduced.

11.4.8 Role of the Opposition

11.4.8.1 The opposition’s task is to convince the judges that the government’s views are fallacious. If the case presented by the government is a truism or requires specific knowledge to debate, the opposition can point this out in their first speech and will be rewarded accordingly by the judges. If the point is well taken, then the opposition must redefine the resolution in a debatable manner.



11.4.8.2 If the government presents a plan case, the opposition can attack the principle, the contention or the arguments for the plan. The opposition can also show how the plan will not work or identify the undesirable side effects that it will create.

11.4.8.3 Finally, the opposition may propose a counter plan that is more effective than the original plan. The first speaker of the opposition must introduce counter plans and the second speaker of the opposition may not add on to the counterplan.

11.4.9 Rebuttals

11.4.9.1 In the rebuttals, the most prevalent elements of the debate must be summarized in a concise and convincing manner.

11.4.9.2 Excluding the first rebuttal of the opposition, no new arguments or facts may be presented unless they directly refute what has already been discussed.

11.4.10 Questions (Points of Information)

11.4.10.1 Questions, also known as Points of Information or POIs, are a secondary means of refuting arguments. They can promptly point out deficiencies in ideas, put speakers on the spot and hog time.

11.4.10.2 The debater who currently holds the floor has the authority to take or ignore questions. The opponent who wishes to ask a question indicates their desire by simply standing up and extending an arm forward.

a. If the debater does not wish to take the question, he or she can indicate by a wave of the hand or a simple "no, thank you". If this occurs, the inquiring party must sit down.
b. If the current debater wishes to entertain the question, it must be stated by the opponent in less than fifteen (15) seconds during which the speaker must yield the floor.

11.4.10.3 The debater who holds the floor, or their team member, may take part in answering the question. The time used to ask and answer the question comes out of the current debater's allotted time.

11.4.10.4 Each debater must accept one question during their speech, if the opponent is making an honest attempt to ask questions.

a. Three attempts at asking a question in one speech constitutes an honest attempt.

11.4.10.5 Questions will not be allowed in the first or last thirty seconds of a speech, or during rebuttal speeches.

11.4.11 Heckling

11.4.11.1 Often, the speaker can contradict himself/herself or make an absurd assumption. Heckling at this time will point out the error and add to the debate.

11.4.11.2 Heckling is acceptable if it is short, to the point and preferably witty. If excess heckling becomes disturbing, the Moderator may intervene.

11.4.12 Points of Procedure

11.4.12.1 If a team believes that one of the rules of the debate have been broken, they must immediately alert the Moderator of the violation by standing and saying, "Point of Procedure."

11.4.12.2 The Moderator will then respond with, "Make your point." The team will then proceed to explain how the debate rules have not been followed.

11.4.12.3 Finally, the Moderator will rule on the point by saying, "Point well taken" or "Point not taken."

11.4.12.4 The time it takes to rise and rule on a point is not included in the speaking time of the team currently debating.

11.4.12.5 The following are violations that warrant a point of procedure:

a. Unprofessional behavior;
b. Offensive behavior;
c. Misquotations;
d. Speaking to the opposition instead of the audience;
e. Presenting new arguments in the rebuttal; and
f. Introduction of parts of a plan by the second speaker.

11.4.12.6 If the government has presented a truism or specific knowledge case, the opposition must wait until the end of the first debater's speech to point this out. Moreover, if the officials have failed to follow the rules of debate (incorrect speaking order, too much speaking time allowed, etc.), it may be politely pointed out to them with a Point of Procedure.

11.4.12.7 Points of Procedures are commonly seen to detract from a debate when offered incorrectly. Points of Procedure which take away from the flow of a debate unnecessarily will be penalized by the judges.

11.4 Procedure / Timeline

11.5.1 Competition Format

11.5.1.1 The structure of Parliamentary Debate will depend on the number of teams. The format is at the discretion of the VP Competition(s) and will be conveyed at the beginning of Day 2.

1.5.1.2 Teams may have to debate two (2) or more times in a row, due to the dynamic nature of the format, however the VP Competition(s) should do their best to ensure teams are given reasonable breaks between debates.

11.5.1.3 Possible formats could include:

1) Round Robin

a. The debate competition could follow a round robin style, followed by a knockout bracket.
b. This will allow all teams to participate in more debates, and to develop a ranking prior to the knockout stage.

2) Double Elimination

a. The debate competition could follow a double elimination format which ensures all teams will get to participate in at least two (2) debates.
b. If a team proceeds unbeaten until the final round, only to lose to another team, there will not be an additional final debate.

11.4 Judging & Rubric

11.6.1 Judging

11.6.1.1 The debate is judged based upon the most convincing argument, communication skills, and ability to follow outlined procedures. Each judge will decide individually, and the Moderator will tally the decisions and announce the winner.

11.6.1.2 The winning team will then move to the next round.

11.6.1.3 Judging feedback shall be provided to each team following the announcement of the winner of that round of debate.

11.6.2 Additional Penalties

11.6.3.1 Exceeding time limit, five (5) point deduction for every 30 seconds.

11.6.3.2 Offensive/unprofessional towards the other team, officials or audience, twenty five (25) point deduction.

11.6.3.3 Misquotations, five (5) point deduction per occurrence.

11.6.3.4 Not following tournament format, fifteen (15) point deduction.